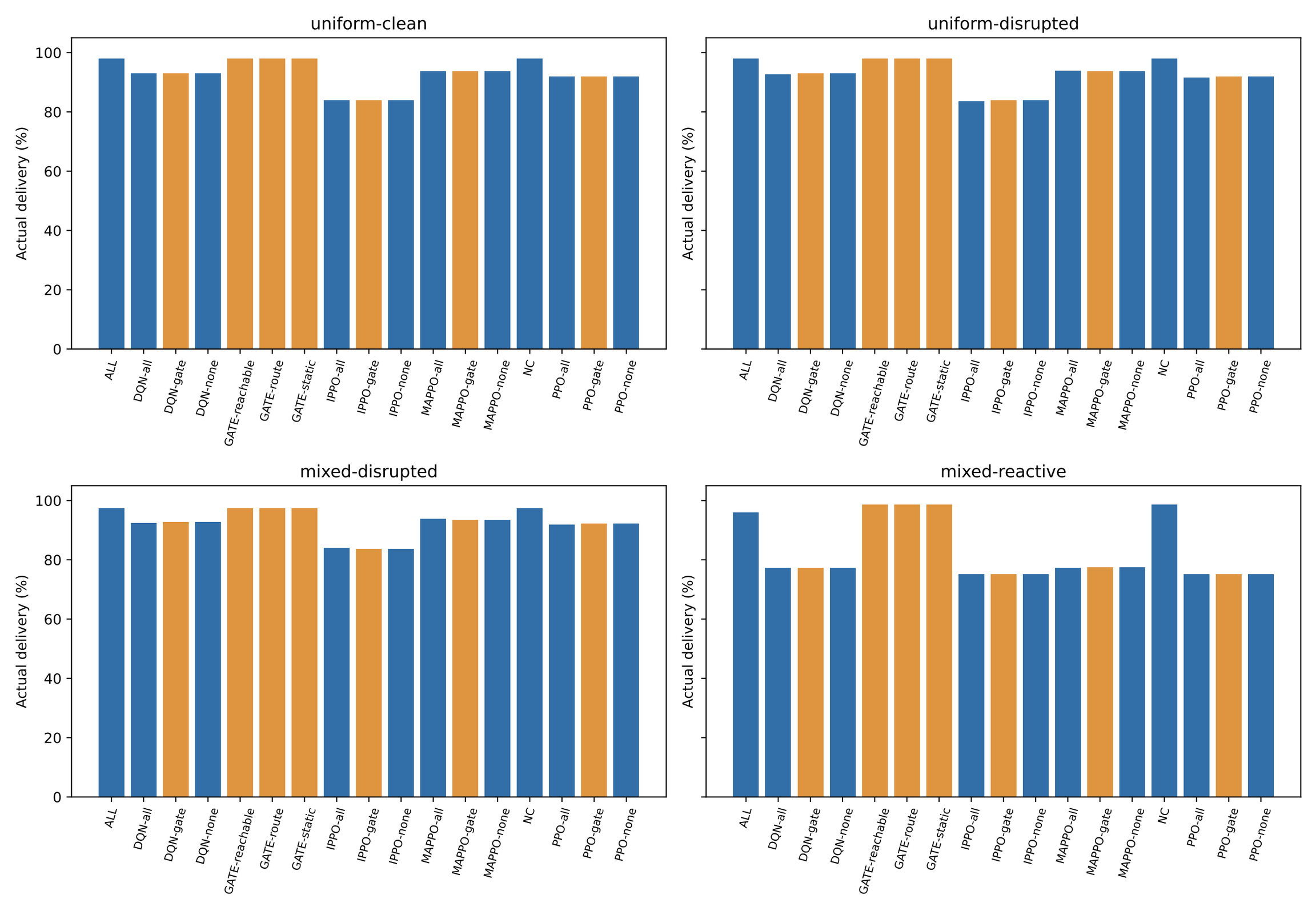
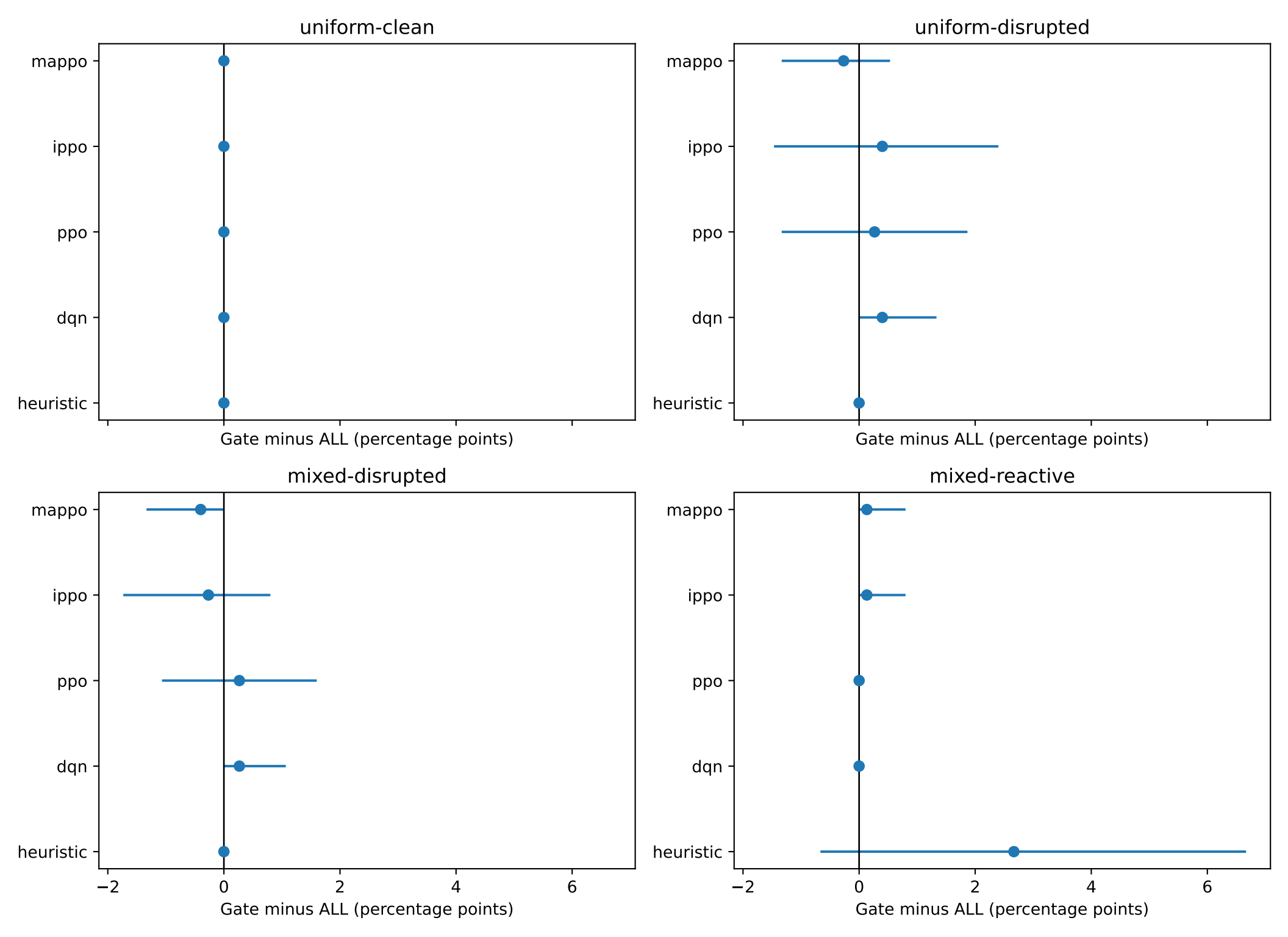
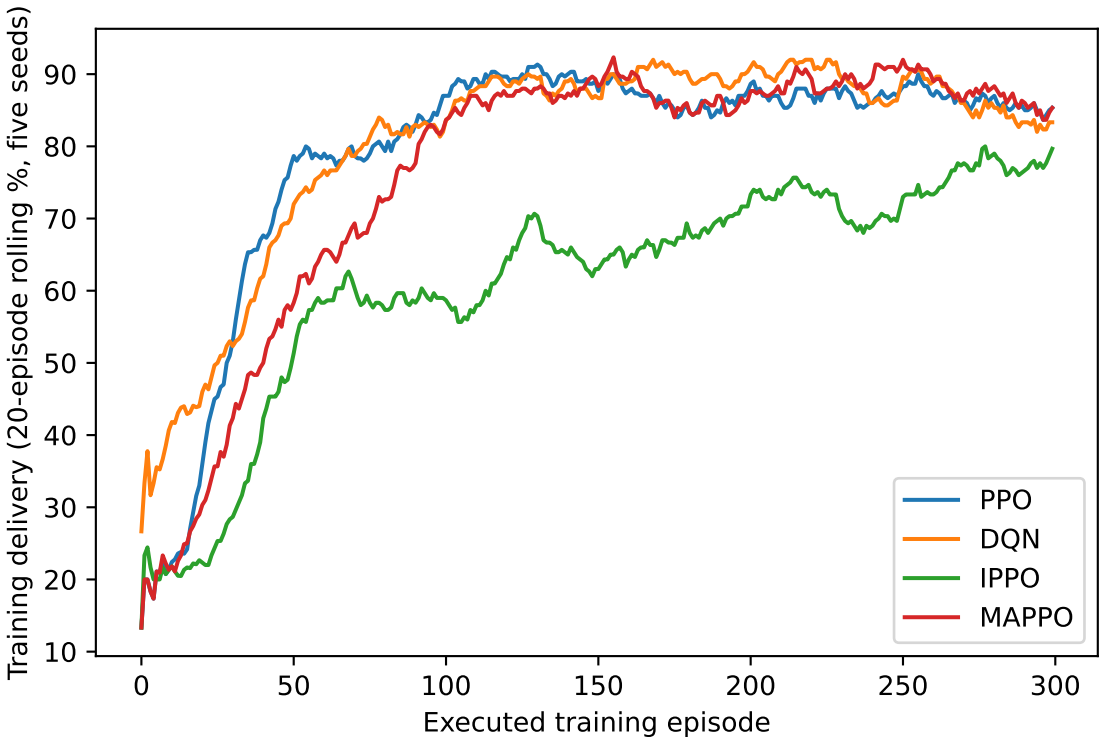
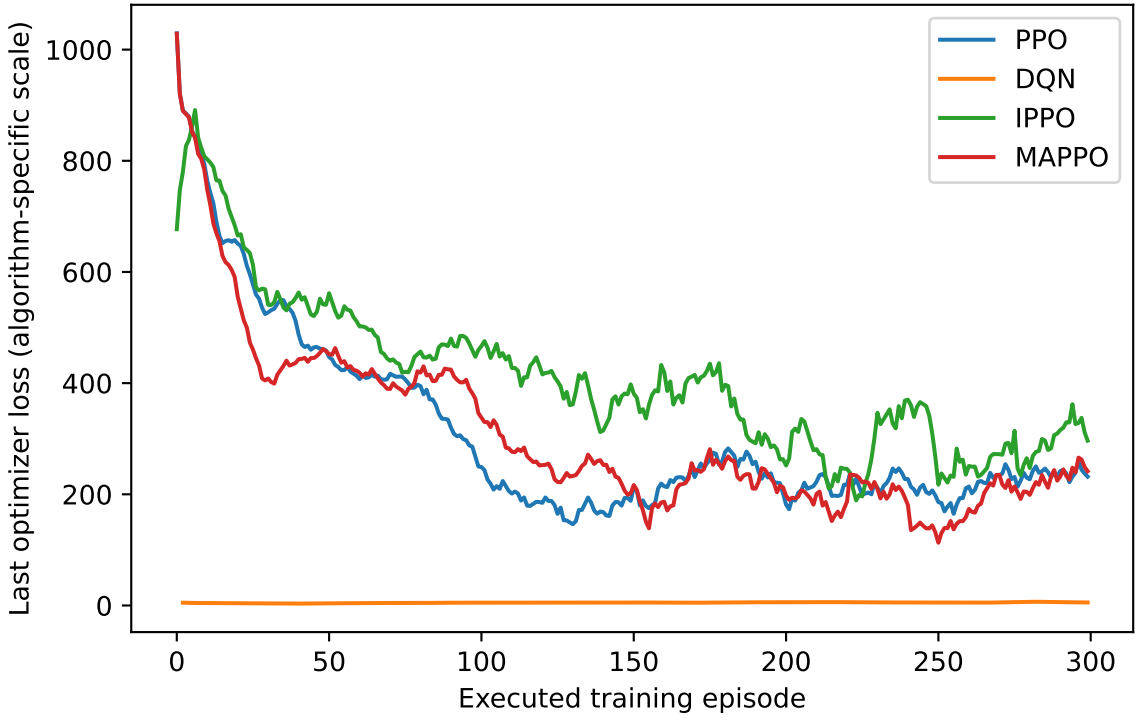
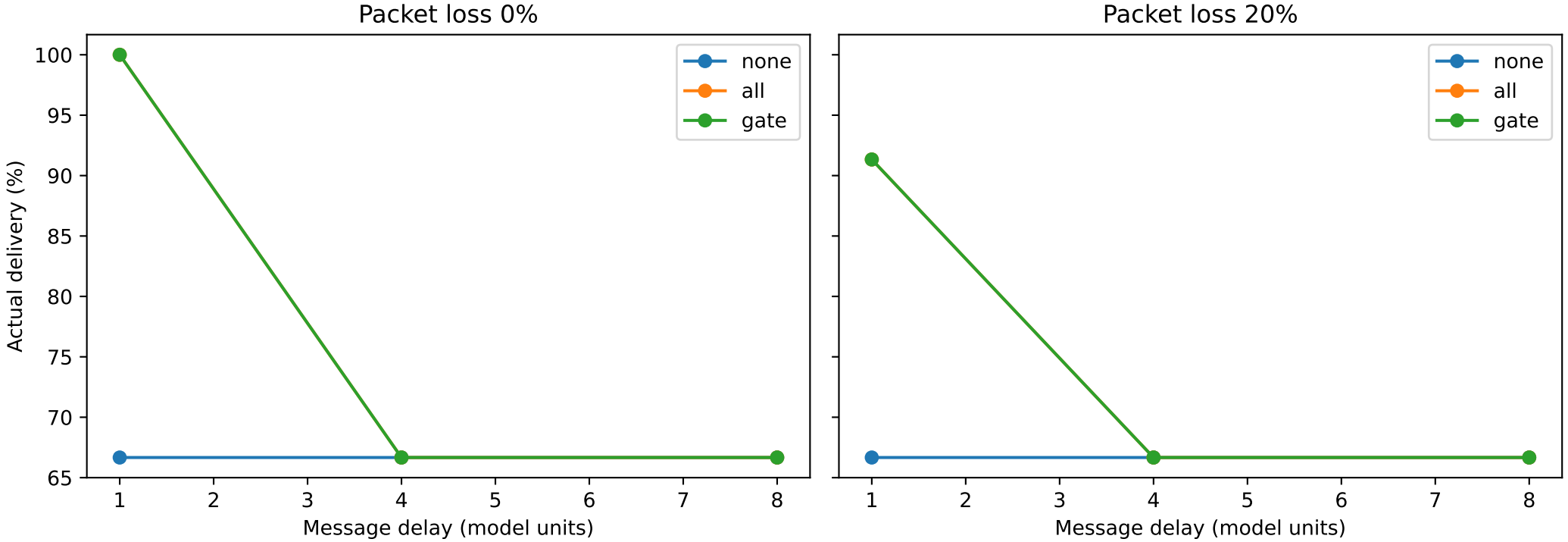
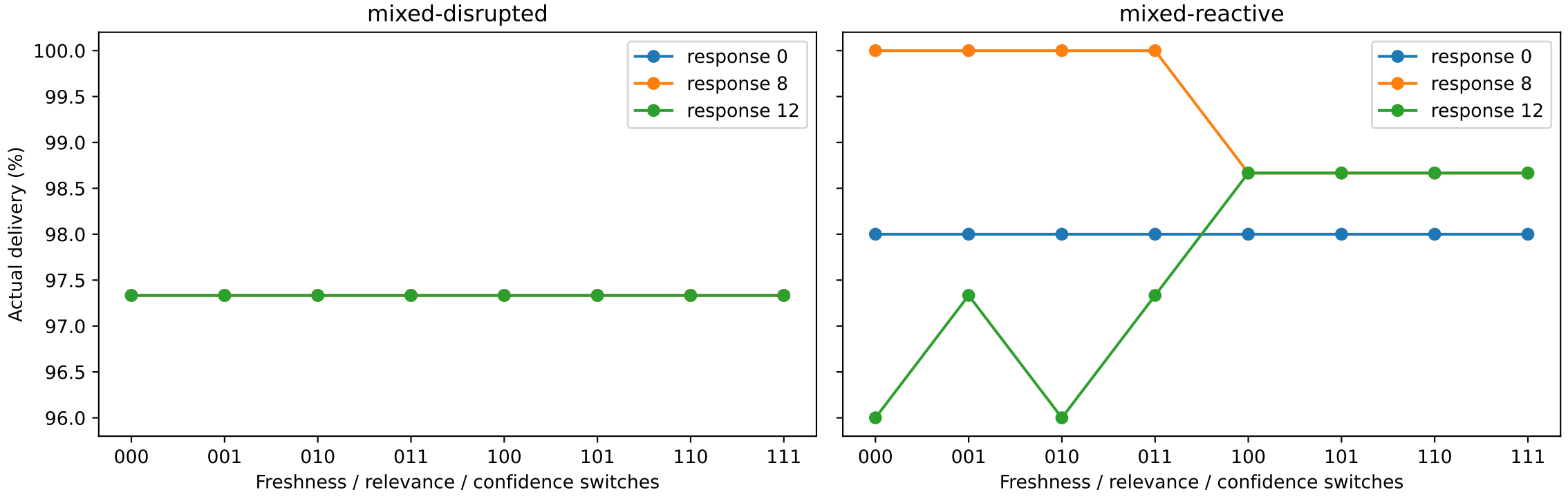

uniform-clean
uniform-disrupted
mixed-disrupted
mixed-reactive
Actual delivery (%)
0
20
40
60
80
100
ALL
DQN-all
DQN-gate
DQN-none
GATE-reachable
GATE-route
GATE-static
IPPO-all
IPPO-gate
IPPO-none
MAPPO-all
MAPPO-gate
MAPPO-none
NC
PPO-all
PPO-gate
PPO-none

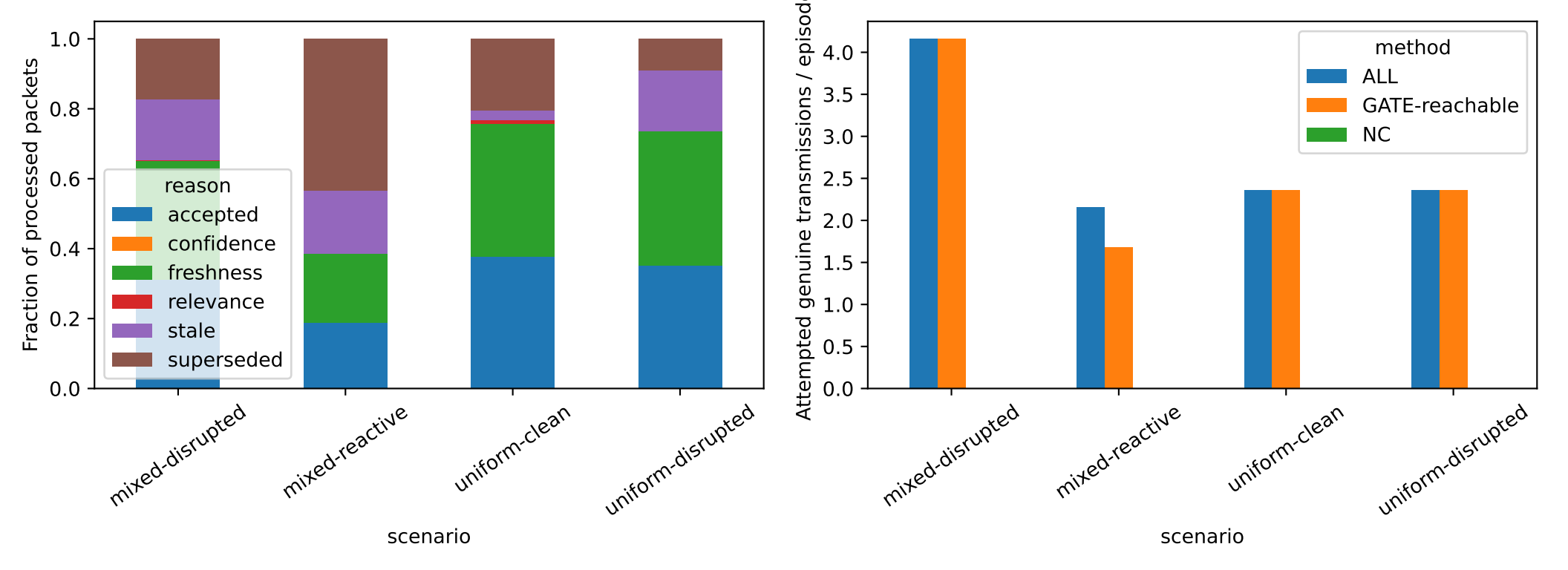

uniform-clean
mappo
ippo
ppo
dqn
heuristic
Gate minus ALL (percentage points)
uniform-disrupted
mappo
ippo
ppo
dqn
heuristic
Gate minus ALL (percentage points)
mixed-disrupted
mappo
ippo
ppo
dqn
heuristic
−2
0
2
4
6
Gate minus ALL (percentage points)
mixed-reactive
mappo
ippo
ppo
dqn
heuristic
−2
0
2
4
6
Gate minus ALL (percentage points)

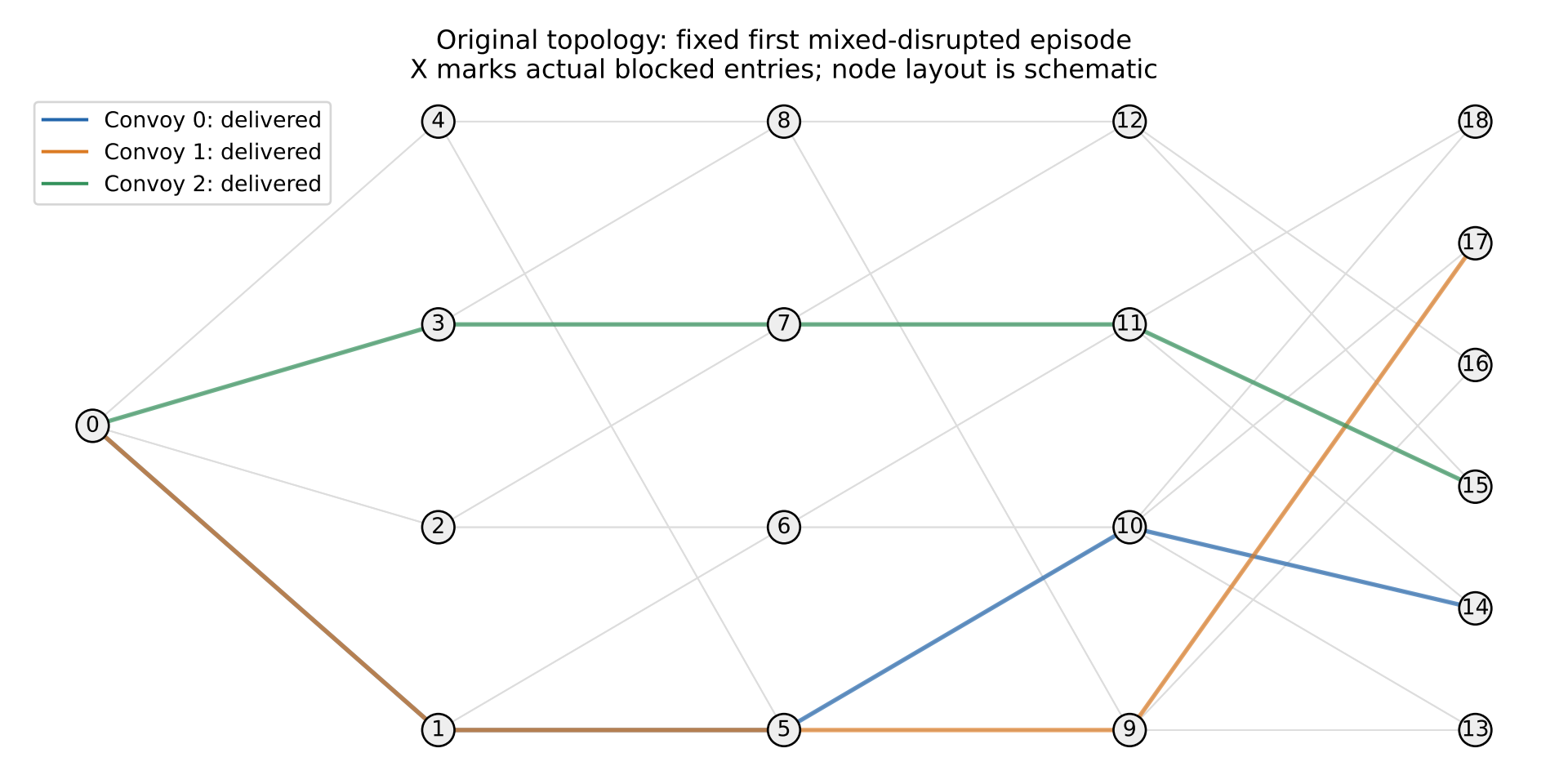

Training delivery (20-episode rolling %, five seeds)
10
20
30
40
50
60
70
80
90
0
50
100
150
200
250
300
Executed training episode
PPO
DQN
IPPO
MAPPO

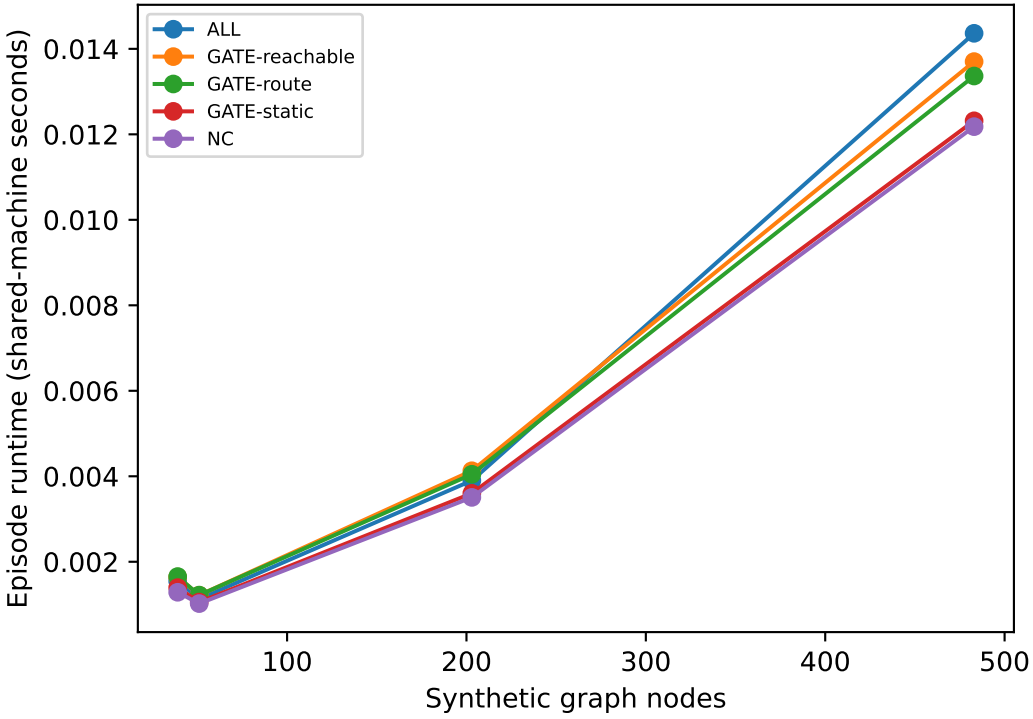

Last optimizer loss (algorithm-specific scale)
0
200
400
600
800
1000
0
50
100
150
200
250
300
Executed training episode
PPO
DQN
IPPO
MAPPO

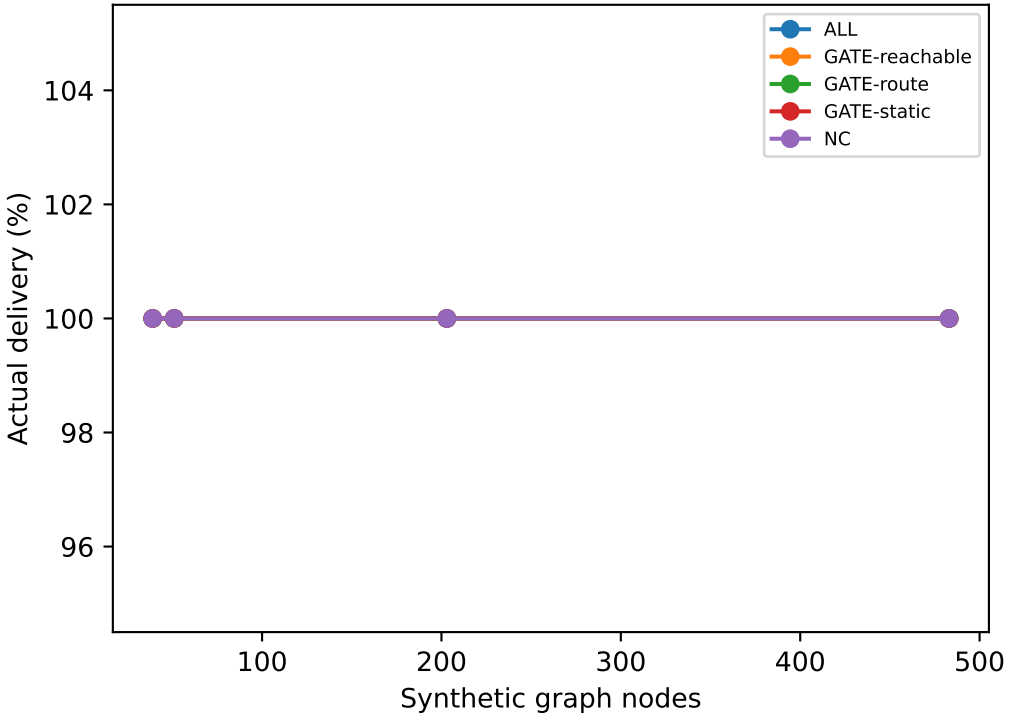

Packet loss 0%
Packet loss 20%
Actual delivery (%)
100
95
90
85
80
75
70
65
1
2
3
4
5
6
7
8
Message delay (model units)
Message delay (model units)
none
all
gate


mixed-disrupted
mixed-reactive
response 0
response 8
response 12
Actual delivery (%)
100.0
99.5
99.0
98.5
98.0
97.5
97.0
96.5
96.0
000
001
010
011
100
101
110
111
Freshness / relevance / confidence switches


Fraction of processed packets
1.0
0.8
0.6
0.4
0.2
0.0
reason
accepted
confidence
freshness
relevance
stale
superseded
mixed-disrupted
mixed-reactive
uniform-clean
uniform-disrupted
scenario
Attempted genuine transmissions / episod
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
method
ALL
GATE-reachable
NC
mixed-disrupted
mixed-reactive
uniform-clean
uniform-disrupted
scenario


Original topology: fixed first mixed-disrupted episode
X marks actual blocked entries; node layout is schematic
Convoy 0: delivered
Convoy 1: delivered
Convoy 2: delivered
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18


ALL
GATE-reachable
GATE-route
GATE-static
NC
Episode runtime (shared-machine seconds)
0.014
0.012
0.010
0.008
0.006
0.004
0.002
100
200
300
400
500
Synthetic graph nodes


ALL
GATE-reachable
GATE-route
GATE-static
NC
Actual delivery (%)
104
102
100
98
96
100
200
300
400
500
Synthetic graph nodes